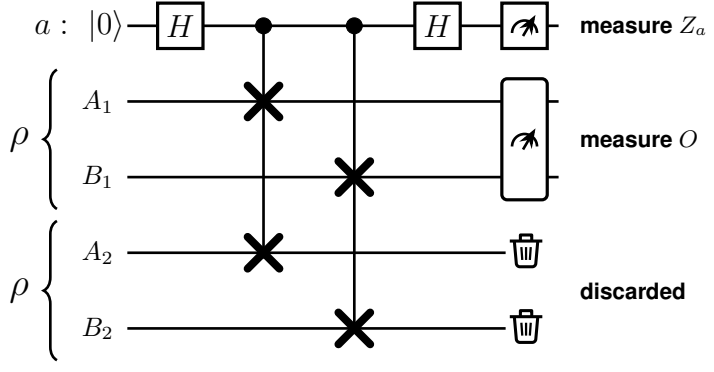


5Q shared ancilla — $\langle Z_a \otimes O\rangle / \langle Z_a \otimes I\rangle = \mathrm{Tr}(O\rho^2)/\,\mathrm{Tr}\,\rho^2$